Type MS · Define a gene mutation and name its causes

Answer 1 of 6 · Question 5 part a.i · 2 marks

2024 NOV · Paper 02 · extract 1 of 2 · find part a.i in this mark scheme extract

5 (a)	(i)	• change in nucleotide/base/codon sequence; • of DNA;		2
	(ii)	• new/different (sequence of) amino acids; • different protein / enzyme / different polypeptide • different phenotype/enzyme does not bind to substrate;		3
(b)	(i)	• allele for haemophilia on X chromosome/ not on Y chromosomes; • hairy ears could only be males / haemophilia could occur in females (and males);		2
	(ii)	• 1 = XX; • 2 = XY^H;		2
	(iii)	• chance of being a girl = 50%/ this can be indicated in a punnet square; • girl will have XX genotype / girl indicated as XX from punnet square/ girls have two X; • no Y chromosome so no chance of hairy ears;		3

Total 12 marks

Type MS · Define a gene mutation and name its causes

Answer 1 of 6 · Question 5 part a.i · 2 marks

2024 NOV · Paper 02 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Define a gene mutation and name its causes

Answer 2 of 6 · Question 6 part e · 2 marks

2019 JUN · Paper 01 (regional R) · extract 1 of 2 · find part e in this mark scheme extract

6(a)	• holds DNA/chromosomes/genetic information in nucleus; • controls what enters and leaves nucleus / allows movement of substances/named substance out of/into the nucleus/exchange of substances between nucleus and cytoplasm;		**(2)**

Question Number	Answer	Additional guidance	Mark
6(b)	B CCG;		**(1)**

Question Number	Answer	Additional guidance	Mark
6(c)	threonine;		**(1)**
Question Number	Answer	Additional guidance	Mark
6(d)	Any six from: • tRNA has anticodons; • translation; • of mRNA; • amino acids carried by tRNA; • joined together; • by peptide bonds; • forming polypeptide/protein; • (final) protein released;	Mp3 must be in context of translation	**(6)**

Question Number	Answer	Additional guidance	Mark
6(e)	• causes change in base sequence/genetic code/named type of mutation; • of DNA;		**(2)**

Total 12 Marks

Type MS · Define a gene mutation and name its causes

Answer 2 of 6 · Question 6 part e · 2 marks

2019 JUN · Paper 01 (regional R) · extract 2 of 2 · find part e in this mark scheme extract

Paper 1: Subject Human Biology

Question Number	Answer	Additional guidance	Mark

Type MS · Define a gene mutation and name its causes

Answer 3 of 6 · Question 8 part a · 3 marks

2020 NOV · Paper 02 · extract 1 of 2 · find part a in this mark scheme extract

8 (a)	any three from • change in base sequence/named mutation (1) • of DNA (1) • causing change in genotype/gene (1) • incorrect protein produced (1) • change in phenotype/characteristics (1)	Allow change in DNA for mps 1 and 2 (1)	3
(b) (i)	father X^HY mother X^HX^h (1) gametes X^H Y X^H X^h (1) F1 X^HX^H normal female; X^HX^h carrier female; X^HY normal male; X^hY haemophiliac male (1)		4
(ii)	(chance of first and second child is) ¼/0.25/25% (1) for both is ¼ x ¼/0.25 x 0.25 (1) = 1/16/0.0625/6.25% (1)		3
(c)	any five from • males only have one X chromosome (1) • two X chromosomes/H and h needed to be a carrier (1) • recessive allele/h (1) • carried on X chromosome (1) • either present or not (1) • no chance for dominant allele to be present (1) • to mask recessive (1)		5

Total 15 marks

Type MS · Define a gene mutation and name its causes

Answer 3 of 6 · Question 8 part a · 3 marks

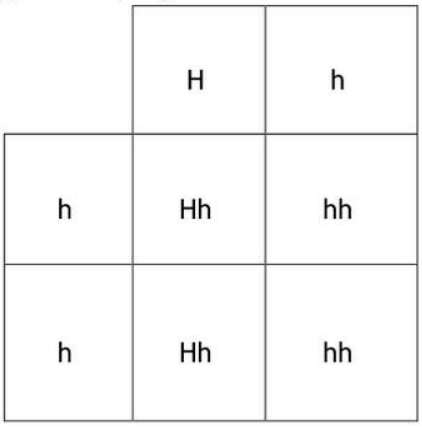
2020 NOV · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract

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Type MS · Define a gene mutation and name its causes

Answer 4 of 6 · Question 3 part a · 1 mark

2024 JUN · Paper 02 (regional R) · find part a in this mark scheme extract

3 (a)	change in the order of bases in DNA;	Allow a description of a type of mutation	(1)
(b)	parent gametes; genotypes of offspring;  	Allow genetic cross Also allow different orientation of alleles in Punnett Square	(2)
(c)	**Advantages** • no (costly) visits to hospital; • no inconvenience of dialysis; • improved quality of life/ longer life/long term solution; **Disadvantages** • reference to difficulty in finding a donor; • rejection/reference to anti-rejection medication for life; • reference to risks associated with surgery;	any one from advantages and any one from disadvantages	(2)

Total for Question 3 = 5 marks

Type MS · Define a gene mutation and name its causes

Answer 5 of 6 · Question 5 part b.ii · 1 mark

2023 NOV · Paper 01 · extract 1 of 2 · find part b.ii in this mark scheme extract

Number				
5	(a)	4^3; = 64;	Full marks for correct answer	2
	(b) (i)	• change in order/sequence; • of bases; • different protein may be formed;		3
	(ii)	uv light/λ rays/ionising radiation/X-rays;	accept any suitable example	1
	(iii)	• can't catalyse reaction/slows reaction/activity; • because can't form E-S complexes/substrate can't bind with active site/active site not complementary;		2
	(iv)	doesn't affect working of enzyme/mutation does not change amino acid/doesn't affect structure/shape of enzyme/different coding for same amino acid;		1

Total 9 marks

Type MS · Define a gene mutation and name its causes

Answer 5 of 6 · Question 5 part b.ii · 1 mark

2023 NOV · Paper 01 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Define a gene mutation and name its causes

Answer 6 of 6 · Question 6 part a.ii · 2 marks

2021 NOV · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

Question number				
6 (a)	(i)	• both increase; • males more than females; • fluctuations; accept any two from;		3 marks
	(ii)	• layer of actively dividing cells; • can become uncontrolled; • when exposed to ultraviolet/sun light;		2 marks
(b)	(i)	17 - 48; = 31 per 100 000; $\frac{31}{25}$ = 1.2/1.24 per 100 000;	allow 16.7 - 17 and 47.8 - 48 then ECF	3 marks
	(ii)	20 per 100 000; $\frac{20 \times 7\,000\,000}{100\,000}$; = 1400;	ECF	3 marks

Total 11 marks

Type MS · Define a gene mutation and name its causes

Answer 6 of 6 · Question 6 part a.ii · 2 marks

2021 NOV · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks